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ORIGINAL ARTICLE



Tribo-mechanical characterisation of MoS₂ and H-BN reinforced PA66 composite

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ABSTRACT

This research aims to create a composite that is inexpensive and can be used in mechanical applications. The main aim is to improve the tribological performance of PA66 by reinforcements. Two fillers were utilised in the experimentation because MoS₂ is a remarkably effective solid lubricant in dry conditions, but the results are not the same in high humidity. H-BN is utilised here to get the benefit in both dry and humid environments and to reduce the negative impact of humidity on the tribological performance of composite. Tensile and hardness tests were performed to identify the mechanical properties, while tribological tests on pin-on-disc were performed to analyse the COF and wear rates of materials. SEM, XRD, and DSC characterisations were utilised to analyse the tensile fracture mechanism, crystallinity, and glass-transition temperature of materials, respectively. The result shows that the strength and hardness of the composite increased in a reasonable amount compared to virgin PA66. Wear and COF tests were conducted with three different parameters, and the results showed an improvement in wear rates of composites compared to virgin PA66.

ARTICLE HISTORY

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KEYWORDS

Polymer composite; PA66;
MoS₂; H-BN; COF; wear

1. Introduction

Polymer composite materials are often used in various engineering areas such as aircraft, automobiles, robots, and electronic equipment. The main advantage of composite materials includes a high strength-to-weight ratio which makes them enable to use in many applications where reduction of weight is essential without compromising the mechanical strength of the component. The automobile sector is one of the largest consumers of polymer composites. The reason for this is vehicle weight reduction which promotes fuel efficiency and reduction in the emission of exhaust thereby reducing air pollution. According to surveys and literature, a 20% to 25% reduction in car mass can save 200 to 250 million barrels of crude oil (Wang, Zheng, and Zheng 2011; Rajak et al. 2019; Pilipović et al. 2020). Polymers and polymer composites' application in the automobile sector includes tires, belts, hoses, leaf springs, driveshafts, bumpers, wear pads, body panels, engine parts, fuel tanks as well as body of the car. In an automobile, wear pads are often made of polymers and polymer composites. These wear pads are light weight, robust, have low cost, and are easily replaceable (Hendre and Bachchhav 2021; Naranje et al. 2021; Hendre and Bachchhav 2019). Polymer composites' application in aircraft includes breaks, window frames, rotors, aircraft wing boxes, blades, vertical fins, tail assemblies, and food tray arms. Weight reduction is very essential in aircraft since it affects fuel efficiency, flying range,

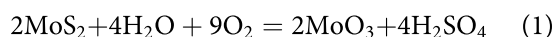
speed, and manoeuvrability (Clyne and Hull 2019). Hybrid composites containing fibres and particulates as reinforcements are also extensively used in various components of aircraft.

Various micro and nanoparticulate reinforcements are frequently used in polyamides to improve wear characteristics. Al₂O₃, SiO₂, wollastonite, kaolin, talc, clay, carbon nanotubes, SiC are found useful in improving wear characteristics and some of the mechanical properties of PA6/PA66 (Guglani and Gupta 2017; Zheng et al. 2009; Hasan et al. 2006; Unal, Mimaroglu, and Alkan 2004; Mohanty and Nayak 2007; Singh et al. 2017; Jogi et al. 2014). Various other oxides, sulphides, and fluoride compounds of metals and non-metals, such as TiO₂, B₂O₃, ZnS, PbS, CaS, ZnF₂, and CaF₂, are found effective in reducing friction and wear rates of polyamides in previously available research (Zhao, Zheng, and Zhao 2006; Bahadur, Gong, and Anderegge 1996; Randhawa and Patel 2020a; You et al. 2014). According to the study of S. Debnath and his team, an increase in the bonding strength between fillers and matrix will not improve the mechanical properties of particulate reinforced composites as it does in fibres reinforced composites (Debnath et al. 2004). Also, the wear rates of materials, in general, are related to the ratio of the indentation hardness (H) to the elastic modulus (E). The lower the ratio H/E, the greater the rate of wear of materials. Fibres and particle reinforcements are advantageous in reducing wear rates in dry conditions rubbing with a smooth surface, while in the

case of abrasive wear, these reinforcements generally increase it. One research stated that the carbon, graphite, MoS₂, polytetrafluoroethylene (PTFE), and short glass fibres are useful in improving the abrasive wear resistance of polymers when embedded in it in optimum amounts (Bijwe et al., 1990). One study revealed that the 3 wt.% of neodymium oxide addition enhanced the microhardness of polymer by 17% and resulted in lower abrasion (Singh and Sharma 2019).

Self-lubricated polymer composites are in trend to avoid external lubrication due to several reasons like contamination, degradation of mechanical properties, and swelling of part. Self-lubrication property is advantageous where one cannot use regular liquid lubrication mainly in the case of polymers. Solid lubricants function in the same way as they develop a low shear strength layer that can shear easily between two surfaces and avoid direct contact between surfaces (Hemanth, Suresha, and Hemanth 2019). MoS₂ and h-BN are widely used solid lubricants in various applications to improve friction and wear characteristics of mating parts.

Molybdenum disulphide (MoS₂) is having a wide service temperature range and high load-carrying capacity. Fretting wear tests with a large range of displacement amplitudes were carried out using MoS₂ bonded solid lubricant coatings. Experimental results showed that such MoS₂ coatings greatly influenced the fretting regimes and reduced the coefficient of friction (Luo et al. 2011). The multi-arc ion plating first deposited a layer of MoS₂ with 2–3 µm thickness, and then the layer of MoS₂ was treated by the low-temperature ion sulphuration to obtain the composite MoS₂ film in one experimentation. The friction and wear result showed that the MoS₂ film possessed an excellent tribological performance by reducing friction (Wang et al. 2007). According to one study, there was oxidation of the MoS₂ to MoO₃ in a moist environment which prevents the flow properties by increasing interparticle cohesion. An increase in the film density of MoS₂ about 7–8 times was found when humidity was increased to 85%. The thermodynamic oxidation route of MoS₂ is shown in equation (1).



The friction coefficient and wear rates of MoS₂ were found to increase as the humidity in the environment increased (Salomon, De Gee, and Zaat 1964; Stewart and Fleischauer 1982; Fleischauer 1984). More quickly oxidation of MoS₂ surfaces, which was facilitated by the induced shear stresses, was found the probable reason for increasing friction and wear rates of MoS₂ in humid environments (Fusaro 1982; Pardee 1972; Prasad, McDevitt, and Zabinski 1999).

Boron-based compounds, such as h-BN, is having excellent friction-reducing properties, while the other compound, i.e., boron carbide, is an excellent wear-

resistant material (Randhawa and Patel 2020b, 2021a, 2021b; Vyas and Randhawa 2021; Mistry and Randhawa 2020). H-BN, also known as white graphite, has a good combination of thermal, thermo-physical, mechanical, and tribological properties (Panda, Bijwe, and Pandey 2019). The tribo-oxidation of h-BN in the presence of water leads to the formation of boron oxide. Low shear strength and viscosity of the lubricious, amorphous boron oxide led to lower friction and wear rates in humid environments.

One of the most promising options for controlling friction and wear in energy-efficient modern systems is the development of self-lubricating composites. MoS₂ is an excellent solid lubricant in dry environments, but as the humidity values in the surrounding atmosphere increase, lubrication properties may reverse and found the opposite, increasing the friction and wear rates due to the formation of MoO₃. H-BN is also a prominent solid lubricant that can work satisfactorily in both dry and humid environments. The effect of MoS₂ and h-BN fillers on the mechanical and tribological properties of PA66 is investigated in this research. H-BN fillers were utilised in the research to nullify the opposite effect of MoS₂ in humid environments. The mechanical properties of PA66 and PA66/MoS₂/h-BN composite was determined by measuring tensile strength, modulus of elasticity, % elongation, and Rockwell hardness. Whereas material's tribological properties, i.e., coefficient of friction (COF) and wear rates were measured by using a pin-on-disc tribometer in this research.

2. Materials and methods

2.1. Materials

Polyamide 66 (PA66) was utilised in the form of granules, while the fillers were in the form of micro-particles. The specifications of the utilised materials are shown in Table 1.

2.2. Samples preparation

PA66 was preheated in an oven at 80°C for 1 hour to remove any moisture from it. After that, granules of virgin PA66 were fed into an injection moulding machine with the help of a hopper. The temperatures were adjusted at 250°C, 255°C, 260°C, and 265°C for zone 1 to 4 of the injection moulding machine. The whole process was carried out at constant pressure and at a low feed rate. After injection moulding, the

Table 1. Specifications of the materials.

	PA66	MoS ₂	h-BN
Size	3–5 mm	< 10 µm (APS)	< 15 µm (APS)
Molecular Weight (g/mol)	224.3	160.07	24.82
Density (g/cm ³)	1.31	5.06	2.1
Purity (%)	99	98	99.98

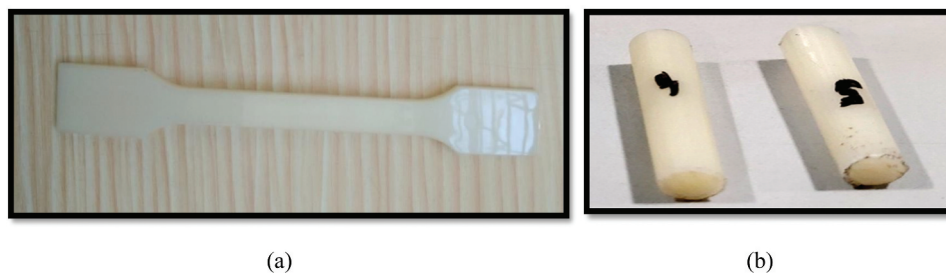


Figure 1. PA66 specimens (a) Tensile test (b) Tribological test.

samples were cooled down to room temperature. Figure 1 shows the injection moulded tensile and tribological test specimens of PA66.

5 wt.% h-BN and 5 wt.% MoS₂ were utilised in the compounding process, i.e., in a 10 kg batch, there were 425 grams of h-BN, 425 grams of MoS₂, and the rest of 8.5 kg was PA66. Compounding of materials was carried out with the help of a twin-screw extruder. The eight different temperatures zones of an extruder were adjusted at 210°C, 220°C, 235°C, 245°C, 220°C, 250°C, 282°C, and 266°C. The rpm of the extruder was adjusted to 444 rpm with a low feed rate. After extrusion, the granules of PA66 composite were heated at 100°C for 30 minutes to remove any moisture from it. The granules of the composite were then fed into the injection moulding machine with the help of a hopper. The temperature profile of four different zones of an injection moulding machine was adjusted at 275°C, 280°C, 285°C, and 290°C for the moulding of composite specimens. After injection moulding, the samples were cooled down to room temperatures. Figure 2 shows the injection moulded specimens of PA66 composite. According to the ISO 527, the width of the narrow segment and the total length of the tensile specimens were kept 21 mm and 170 mm, respectively. Pin specimens of tribological testing were having the specifications of 40 mm length and 4 mm diameter (ASTM G99).

2.3. Mechanical and tribology testing

Tensile tests were performed at PI Polyblendz Pvt. Ltd., Gujarat, India. A universal testing machine (UTM) was utilised to determine the tensile strength, elongation, and modulus of elasticity of materials. A Rockwell hardness tester was utilised to measure

the hardness of virgin PA66 and PA66 composite. Tribological tests were carried out using a DUCOM TR-20LE-200PHM pin-on-disk tribometer and according to the standard ASTM G99. In the tribological testing, no external lubricant was used, and all experiments were conducted at a room temperature of 27°C. The hardened EN31 steel disc was utilised in the tribological testing as the other mating and rubbing surface. The 100 mm track diameter was selected for testing on the tribometer. RPMs were selected as 300, 600, and 900 and the loads were 0.5 kg (4.9 N), 1 kg (9.8 N), and 1.5 kg (14.71 N). All tests were performed for a 1000-metre run, and the duration of the tests were calculated from equation (2) and (3) and were 11 mins, 6 mins, and 4 mins for 300, 600, and 900 rpm, respectively.

$$V = (\pi * D * N) / 1000 \text{ m/min} \quad (2)$$

$$T = 1000 / V \quad (3)$$

3. Results and discussion

3.1. Mechanical testing

Table 2 presents the tensile test results of virgin PA66. The average tensile strength was found 76.08 MPa after 5 samples experimentation. The average elongation % observed was 27.69%, and the average modulus of elasticity was 1637.03 MPa after 5 samples of experimentation.

Table 3 shows the tensile test results of PA66/MoS₂/h-BN composites. The average tensile strength of PA66 composite was found 86.32 MPa, Elongation 12.24%, and modulus of elasticity 1848.79 MPa after five samples of experimentation.



Figure 2. PA66 composite specimens (a) Tensile test (b) Tribological test.

Table 2. Tensile test results of pure PA66 specimens.

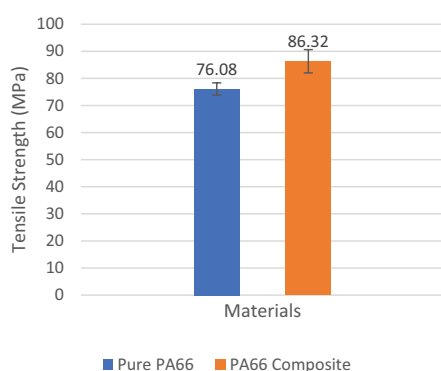
Specimen	Width mm	Thickness mm	Cross-sectional Area mm ²	Max. Load N	Tensile Strength MPa	Elongation %	Modulus of Elasticity MPa
1	21	4	84	2818	70.46	34	1532.006
2	21	4	84	2795	69.89	30.18	1513.2452
3	21	4	84	3203	80.09	25.11	1747.0653
4	21	4	84	3209	80.24	19.34	1710.7567
5	21	4	84	3190	79.75	29.82	1682.0683
Average	21	4	84	3043	76.08	27.69	1637.0283

Table 3. Tensile test results of PA66 composite specimens.

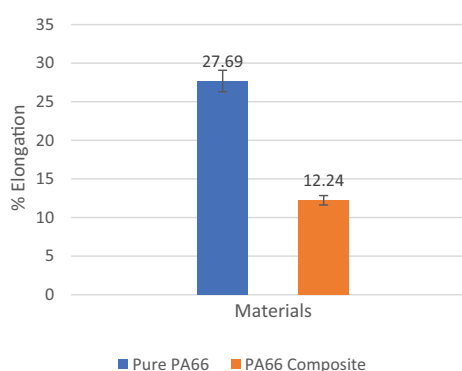
Specimen	Width mm	Thickness mm	Cross-sectional Area mm ²	Max. Load N	Tensile Strength MPa	Elongation %	Modulus of Elasticity MPa
1	21	4	84	3536	88.46	12.92	1893.9175
2	21	4	84	3421	85.53	9.07	1812.771
3	21	4	84	3404	85.15	15.15	1792.798
4	21	4	84	3457	86.44	14.65	1853.1524
5	21	4	84	3441	86.02	9.41	1891.289
Average	21	4	84	3451	86.32	12.24	1848.7856

Figure 3 shows the comparative results of mechanical testing of PA66 and PA66 composite. Virgin PA66 and PA66 composite's average tensile strength was found 76.08 and 86.32 MPa, respectively. The tensile strength increases 13.46% by the addition of MoS₂ and h-BN fillers in the PA66. An increase in tensile strength may be due to the adequate bonding between the reinforcing particles and the matrix. The modulus of elasticity was found 1637.0289 & 1848.7856 MPa

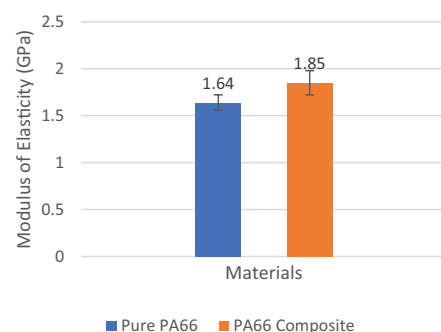
and the % elongation was 27.69% and 12.24% for virgin PA66 and PA66 composite, respectively. The elastic modulus increases 12.8%, and elongation decreases 55.8% for the PA66 composite. The addition of MoS₂ and h-BN fillers in PA66 resulted in brittleness and reduced the elongation of the composite material. The average Rockwell hardness of PA66 and PA66 composite was found 504 and 514, respectively (Table 4). The increase in the hardness of



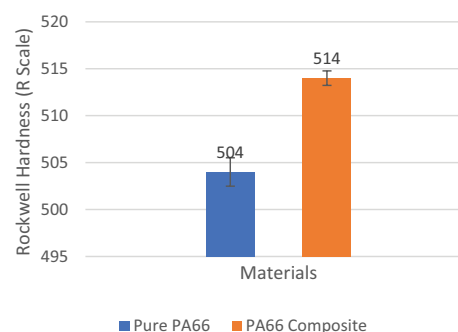
(a)



(c)



(b)



(d)

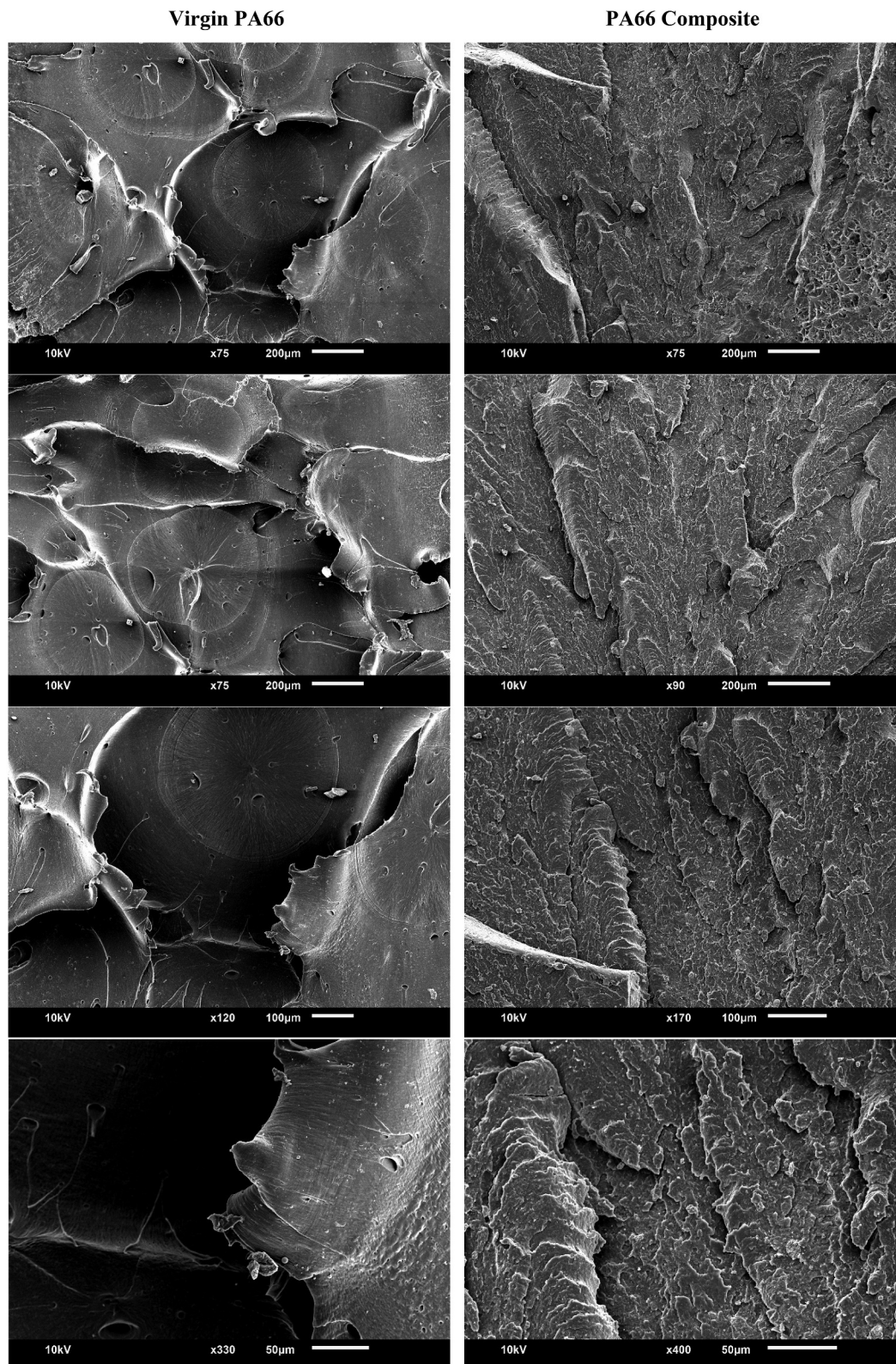
Figure 3. Mechanical properties of PA66 and PA66 composite (a) Tensile strength (b) Modulus of elasticity (c) Elongation (d) Rockwell hardness.

Table 4. Rockwell hardness.

Sample	Pure PA66	PA66 Composite
1	512	516
2	480	500
3	538	511
4	500	527
5	490	516
Average	504	514

composite was almost 2% compared to virgin PA66. The fillers reinforcement increased the tensile strength, elastic modulus, and hardness of PA66 while it reduced the % elongation of PA66 material.

SEM images of the tensile fracture of materials are shown in Figure 4. The figure's left-hand side shows the SEM images of virgin PA66, and the right-hand


Figure 4. SEM images of the tensile fracture.

side shows the SEM images of PA66 composite. The virgin PA66 exhibits slight extension during the tensile fracture of specimens which is clearly visible in the extended cracked portions of PA66's SEM images. In contrast, the PA66 composite experienced a brittle fracture without any extensions which can be seen in the PA66 composite's SEM images. The SEM characterisation relates to the obtained elongation test results in which the PA66 composite samples had less elongations before the tensile fracture compared to virgin PA66 (Akil et al. 2014; Zhou and Lucas 1995).

3.2. Coefficient of friction and wear

Tables 5 and Tables 6 present the COF and wear rates of PA66 and PA66 composite. The data were collected for different rpm, load, and sliding durations. As the rpm and load increased, there was an increase in the COF and wear rates of the materials. As per the obtained data, it can be observed that the COF and wear rates are higher for virgin PA66, and the same was found substantially lower for the PA66 composite. The average of the obtained results was then plotted to see the comparative results of COF and wear rates. The weights of the samples were measured before and after the tribological testing and converted to the percentages by equation (4) for easy plotting of COF and wear rates into the same graph.

$$\text{Wear (\%)} = \frac{\text{Weight of sample before testing} - \text{Weight of sample after testing}}{\text{Weight of sample before testing}} * 100 \quad (4)$$

Table 5. Tribometer testing results of PA66.

	Speed	Weight	Frictional Force (FF)	Coefficient of Friction ($\mu = \text{FF}/\text{Weight}$)	Weight of Pin(gm)		Wear %
	RPM	N	N		Before	After	
1	300	4.9	2.76	0.562	1.127	1.125	0.18
2	300	9.81	3.6	0.367	1.214	1.211	0.24
3	300	14.71	5.73	0.39	1.140	1.137	0.26
4	600	4.9	2.03	0.414	1.218	1.215	0.24
5	600	9.81	3.7	0.377	1.183	1.180	0.25
6	600	14.71	5.9	0.40	1.173	1.169	0.34
7	900	4.9	1.66	0.338	1.188	1.185	0.25
8	900	9.81	5.73	0.584	1.205	1.199	0.49
9	900	14.71	8.06	0.548	1.196	1.186	0.83

Table 6. Tribometer testing results of PA66 composites.

	Speed	Weight	Frictional Force (FF)	Coefficient of Friction ($\mu = \text{FF}/\text{Weight}$)	Weight of Pin(gm)		Wear %
	RPM	N	N		Before	After	
1	300	4.9	1.6	0.326	1.215	1.213	0.16
2	300	9.81	4.73	0.482	1.126	1.124	0.18
3	300	14.71	7.23	0.49	1.268	1.265	0.24
4	600	4.9	3.63	0.74	1.211	1.209	0.165
5	600	9.81	5	0.51	1.125	1.123	0.18
6	600	14.71	9.26	0.63	1.187	1.184	0.25
7	900	4.9	2.93	0.597	1.185	1.183	0.17
8	900	9.81	6.30	0.642	1.215	1.212	0.24
9	900	14.71	8.56	0.58	1.180	1.177	0.25

As shown in Figure 5, the COF of PA66 composite was found higher compared to virgin PA66. That was due to the induced hardness in the composite by MoS₂ and h-BN fillers. Also, it can be seen that the wear rate of the composite was much lower compared to virgin PA66. There was not much effect of lubricating transfer film formation seen on the counter steel disk. In fact, the reduction in wear rates of the composite was due to the increased hardness, which protected the material to worn out. The increase in COF of composite is 25%, while wear rate reduction is 41% compared to virgin PA66.

The wear tracks' SEM images of tribological testing are shown in Figure 6. The causes of wear were stick and slip for the virgin PA66, in which material tends to stick during matting and rubbing with steel disc, and due to continuous movement, it slips (Soundararajan et al. 2019). The temperature was also playing a vital role in this phenomenon. Due to the increase in temperature between polymer pins and steel disc, the material stuck to the disc, and the mechanical force of the rotating steel disc made it slip after sticking. The PA66 composite materials' SEM images revealed the fine tracks with less depth of wear. PA66 composite experienced abrasive wear due to its increased hardness (Jia et al. 2015; Randhawa and Patel 2021c). The asperities or roughness of the steel disc made the PA66 composite's materials ploughed out from it during the continuous movement or rotation of the disc. There was no stick-slip for the composite samples seen on the steel disc; even the formation of lubricating transfer film was either very thin or not present during the testing.

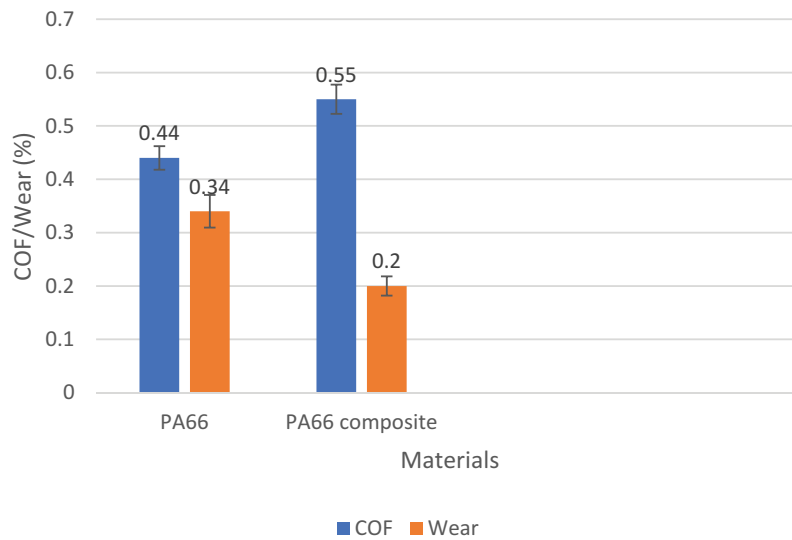


Figure 5. COF and wear of PA66 and PA66 composite.

DSC characterisation of PA66 composite revealed an increase in the material's thermal stability, as shown in Figure 7. The glass transition temperature of the PA66 composite increased significantly, and an increase of 11°C of glass transition temperature was observed for the PA66 composite compared to virgin PA66. MoS₂ and h-BN are thermally stable materials with high glass transition temperatures and melting points and these thermally stable fillers increased the glass transition temperature and thermal stability of PA66 composite to some extent.

The XRD analysis was carried out on a Bruker D2 phaser x-ray diffractometer at a scanning speed of 10° per minute with Cu-Kα radiation ($\lambda = 0.154$ nm) at room temperature. Tests were carried out in a scanning range of 10° to 60° 2 θ . Figure 8 presents the XRD plot of PA66 composite in which the X-axis represents 2 θ values, and the y-axis represents counts per second (CPS) or intensity of the peaks. PA66 is a semi-crystalline polymeric material, so the degree of crystallinity (DOC) of PA66 was found 39.5%, and the rest was the amorphous amount. For the PA66 composite, the DOC was 41.4%, and an increase in

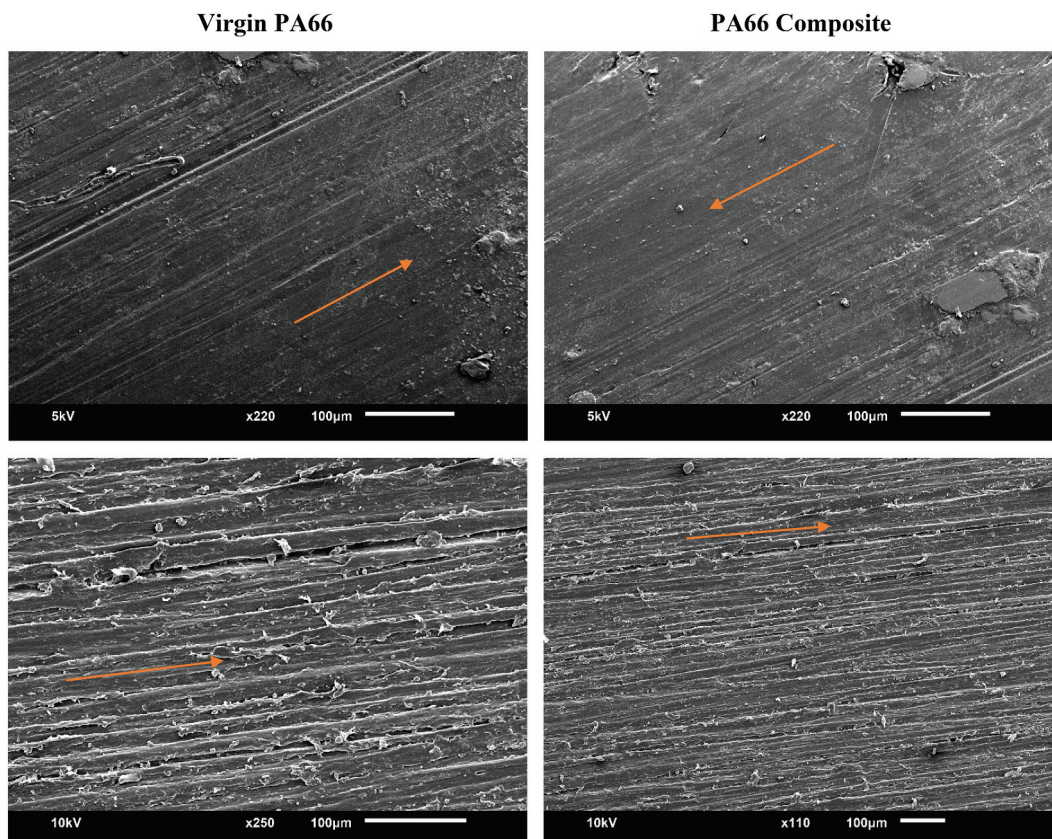


Figure 6. SEM images of wear tracks.

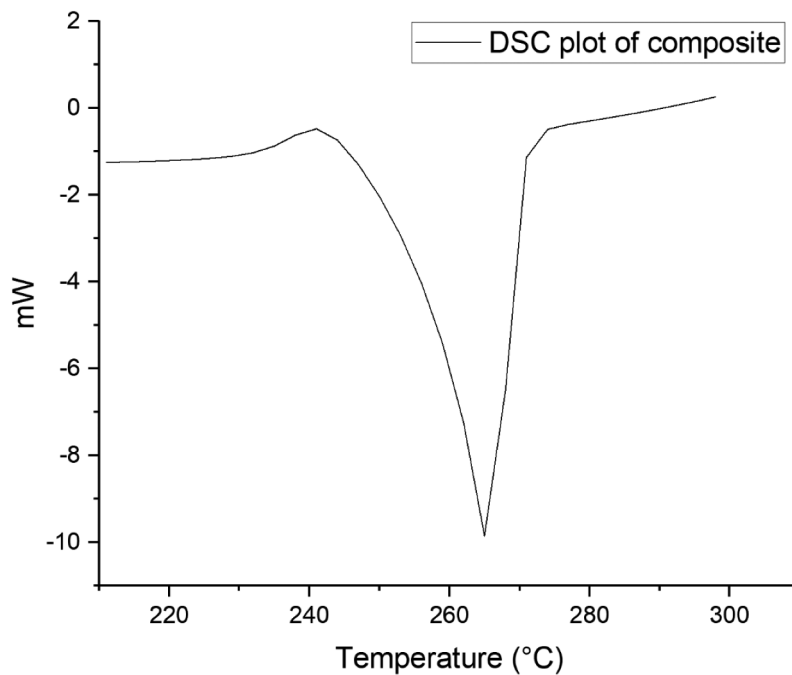


Figure 7. DSC plot of PA66 composite.

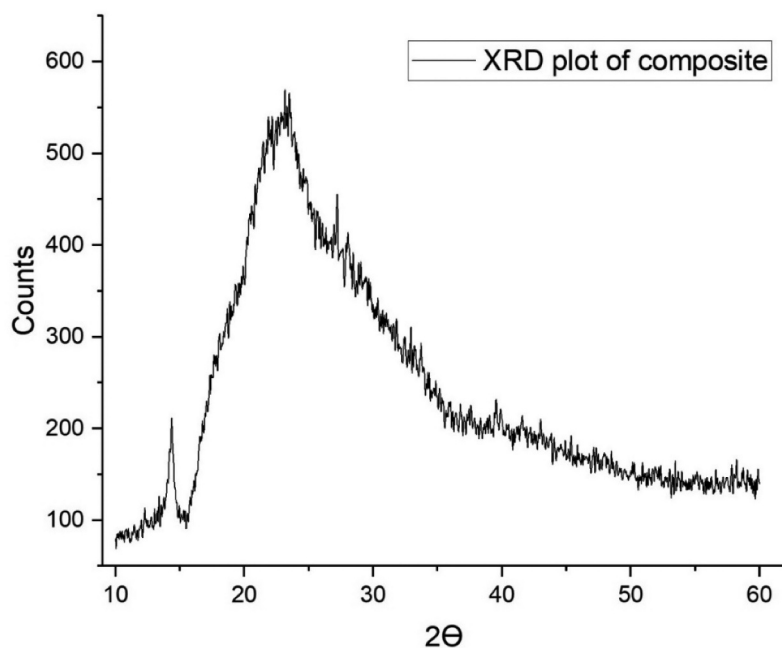


Figure 8. XRD plot of PA66 composite.

crystallinity of the PA66 composite was almost 5%. This increase in crystallinity was one of the reasons for the improved strength, hardness, and wear resistance of the PA66 composite.

4. Conclusion

The influence of MoS₂ and h-BN fillers on the mechanical and tribological properties of PA66 was investigated in this paper. The mechanical properties like tensile strength, elastic modulus, and hardness

were increased while the elongation % was found to decrease for the PA66 composite compared to virgin PA66. The COF of composite was found higher, while the wear rates were found to decrease almost half of virgin PA66. DSC characterisation revealed the increase in glass transition temperature, while XRD analysis revealed the increase in crystallinity of composite compared to pure PA66. The increase in crystallinity was found one of the main reasons for an increase in strength and hardness of the composite. There were either very thin or no transfer film

formations found on the steel disc during the tribological testing of the composite. The decrease in wear rates was purely due to the increase in the hardness of the composite. The MoS₂ and h-BN fillers reinforcement were found very effective in improving mechanical properties and wear resistance of PA66, and the composite can be utilised at the places where PA66 has to undergo frequent rubbing. Also, the PA6/MoS₂/h-BN composite material may be effective in high humidity areas where polyamide components must work and withstand high moisture content in the environment.

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Data availability statement (DAS)

The authors confirm that the data supporting the findings of this study are available within the article.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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